Thermally Reversible Beta-Sheet-like Ordering Controls Gelation in Elastin-like Polypeptides

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Abstract

Amyloid-like β-sheet packing spans a wide range of stabilities, from the kinetically persistent cores of pathological fibrils to contacts that come apart again, and where a sequence falls in that range decides whether it can serve as a switchable material. Here, temperature-dependent small- and wide-angle X-ray scattering of the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$ resolves a reversible transition between a dispersed low-temperature state and a hierarchical high-temperature assembly carrying β-sheet-like peaks at $Q = 1.38$ and 0.72 Å^{-1} that vanish on cooling. Chain length

decides whether the gel returns: $(VPAVG)_{30}$ recovers its low-temperature profile through repeated cycling while the ten-repeat construct does not. The reversible case therefore places amyloid-like local order in the role of a physical junction rather than a terminal aggregate.

Introduction

Amyloid-like β-sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence homology are associated with Alzheimer's disease,[1] Parkinson's disease,[2] type II diabetes,[3] prion diseases,[4] and systemic light-chain amyloidosis.[5] Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.[6] Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules[7] and build the extracellular matrix of bacterial biofilms,[8] and related assemblies are increasingly used as mechanically robust biomaterials.[9] What differs across these systems is not whether β-sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,[10] and where a sequence falls in that range is a materials-design variable.

Reversible β-sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.[10–13] Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.[14–18] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.[17–19] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.[20,21] Although these studies establish the arrested state from time-averaged scattering and rheology, how the assembly evolves toward arrest remains elusive.

Results and Discussion

Figure 1 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were measured at APS 12-ID-B by simultaneous SAXS/WAXS at 13.3 keV.[22–24] The "Measurement" aliquot had undergone more than 10 heating/cooling cycles; the "Reference" aliquot had stayed at 4 °C. At 10 °C the two overlap, and a Guinier fit to the Reference gives an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix), a dispersed state with no mesoscale aggregates down to $Q = 0.0081$ Å^{-1}, which corresponds to structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å^{-1}). Both also develop the same pair of high-Q features, at $Q = 1.38$ and 0.72 Å^{-1} in the Reference and 1.36 and 0.74 Å^{-1} in the Measurement. The cycled aliquot alone makes the point: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at 1 °C min^{-1} (SI Appendix). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on every heating and returned to the same low-temperature state on every cooling (SI Appendix). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle scattering signifies density heterogeneity extending across the probed interval rather than a single resolved structural scale.[25] Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.[20,21] The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross-β-like

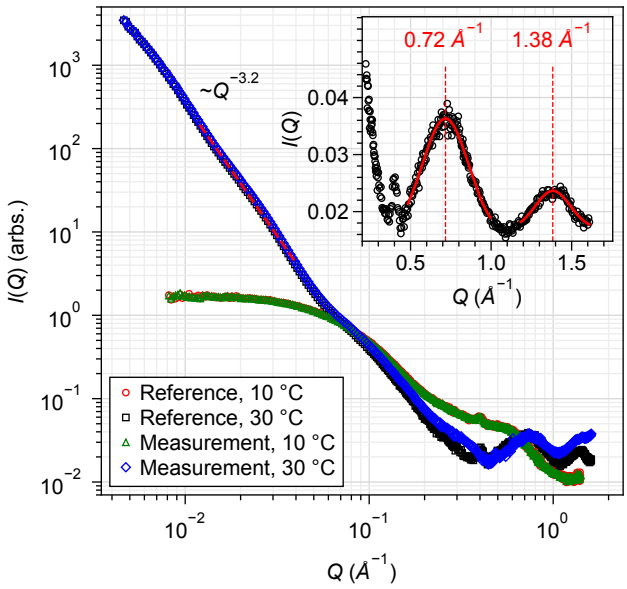


Figure 1: Temperature-dependent structure and reversibility of $(\mathrm{VPAVG})_{30}$. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix). Heating produces an approximate $Q^{-3.2}$ low-Q response and peaks at $Q = 1.38$ and 0.72 Å^{-1}, corresponding to real-space spacings of approximately 4.5 and 8.8 Å, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040$ Å^{-1}. Inset: the high-Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å^{-1}, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.[6,26] Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 $Å^{-1}$ (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross-β crystallites. Their concurrent appearance with the low-Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β-sheet-like associations that are plausible, rather than uniquely established, physical junctions.

The comparison of Figure 1 sets two separate 12-ID-B aliquots against each other. The test in Figure 2 is more direct: one aliquot at 8-ID-I, taken through seven consecutive cycles and compared with itself, so that any difference between cycles can only be irreversibility. The aliquot is a separate loading of the same stock, measured in the 2022-1 beamtime. Each cycle held the sample at 6 °C, ramped to 34 °C at 1.01 °C min^{-1} over 270 consecutive 2 s acquisitions, and returned to 6 °C, taking 5.8 h in total. No acquisitions were taken during cooling, so the cooling segments of Figure 2a are a projection from the fixed control protocol rather than a measurement; every solid segment is measured, once per acquisition, from the RTD of the stage zone holding the cell.

Three states are sampled in every cycle: 6 °C, and two windows near the top of each ramp at 31.9 °C and 33.8 °C. Averaged over $0.004 < Q < 0.008$ $Å^{-1}$, the seven cycles agree to a relative standard deviation of 9% at 6 °C and 5% at 33.8 °C, with no drift from the first cycle to the seventh. At 31.9 °C the spread is 33%, but that reflects where on the ramp each cycle happened to be sampled rather than a difference between cycles: the windows fall between 31.62 °C and 32.48 °C, which is on the steep part of the transition. Regressing the band-averaged $\ln I$ of each cycle on that cycle's mean window temperature gives $\mathrm{d}\ln I/\mathrm{d}T = 0.95\,°C^{-1}$ with $r = 0.974$, and the spread taken about that line falls to 7%. The same regression at 33.8 °C, past the steep part, takes 5% to 1%. The cycles therefore repeat to better than 10% at every state once the sampling temperature is accounted for.

The correlation functions say what state the sample was in when each was taken (Figure 2c). At 33.8 °C the correlation function shows no resolvable decay anywhere in the accessible window, from 10^{-2} to 2 s, in any of the seven cycles: the sample was arrested on every heating. At 31.9 °C it decays throughout that same window, so the assembly is still mobile 1.9 °C lower. The dashed line in that panel is the measured correlation level of a static reference recorded in the same beamtime, read at the same Q bin. A static sample cannot relax, so its level is the instrumental contrast, and showing it as a curve rather than as a fitted number keeps the comparison between two measurements. The 33.8 °C correlation functions sit just below it and the 31.9 °C ones fall well away from it. A contrast value is needed to fit a correlation function, which is why a fitted series carries its own calibration; it is not needed to judge whether a sample has arrested, which is what this panel does, so no arrested fraction is quoted from it. No correlation function is shown at 6 °C, where the coherent signal at the lowest Q is too weak for any contrast to be resolved.

Materials and Methods

To be assembled from the pre-split Letter (tag `pre-split` in PubCode-2021-IDPP): the 12-ID-B paragraph, the 8-ID-I cell and beam description, and the robotic cell-assembly method.

Supporting Information

Recombinant synthesis and purification; sample environments and the automated cell assembly; SAXS/WAXS calibration and merging; Guinier analysis; chain-length ramps; seven-cycle thermal repeatability; irreversibility of the ten-repeat construct; videos of the reversible transition and of the robotic assembly (PDF, MP4).

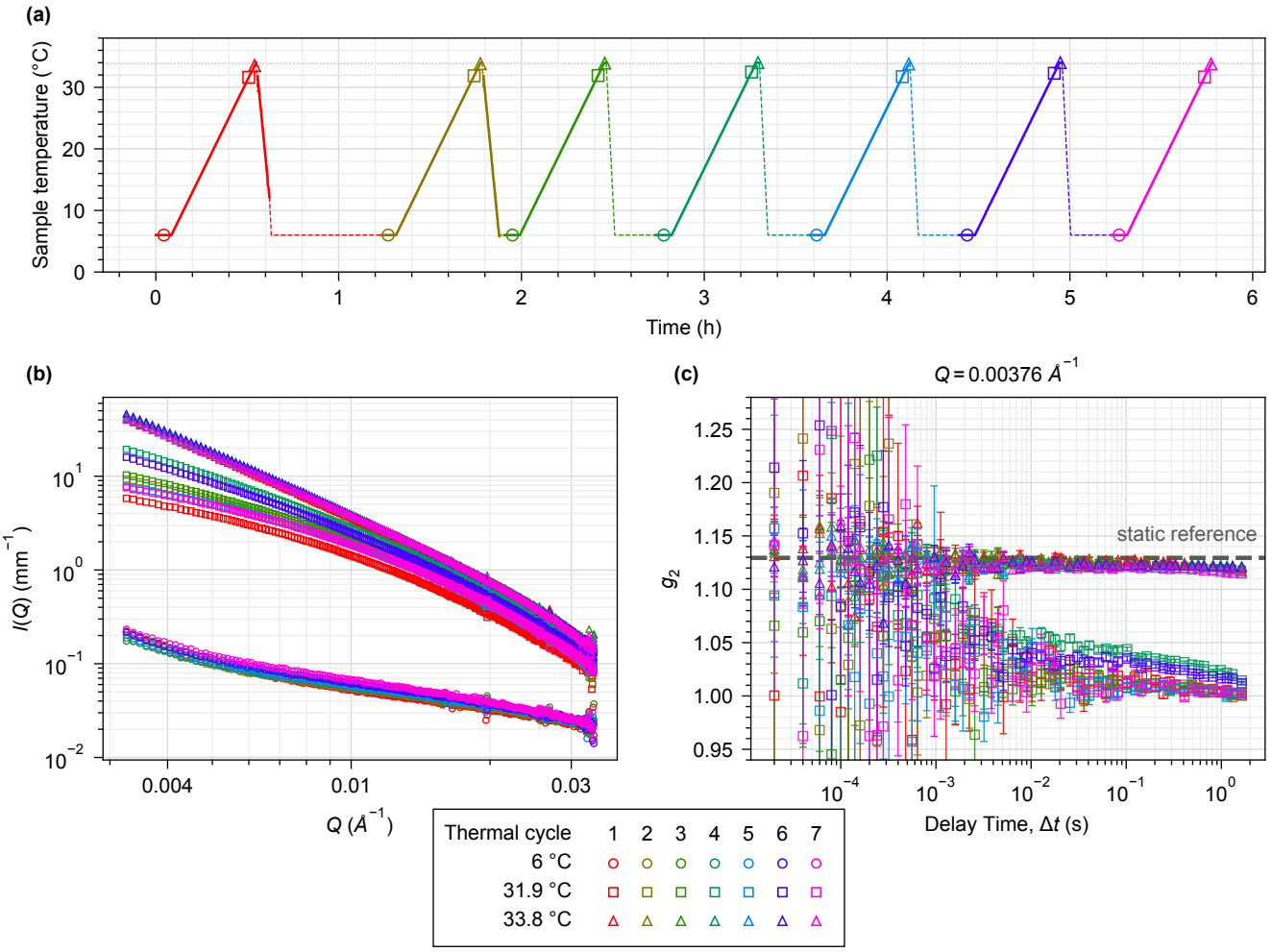


Figure 2: Thermal-cycling repeatability of the SA-XPCS sample: one aliquot through seven consecutive cycles. (a) Sample temperature over the 5.8 h sequence. Solid curves are measured: the stage RTD was read once per 2 s acquisition. No acquisitions were taken while the sample cooled, so no temperature was recorded over those intervals; the dashed curves there are not data but the profile projected from the control protocol, which was fixed at 10 °C min^{-1} cooling until 6 °C followed by a hold at 6 °C until the next ramp. Open symbols mark, at the time and temperature at which it was taken, every acquisition group that is plotted in (b) and (c), so each lower-panel curve can be traced to its point in the thermal history. (b) Buffer-subtracted SAXS on an absolute scale at 6 °C (circles, 32–41 surviving acquisitions averaged; see the text), 31.9 °C (squares, acquisitions 241–250) and 33.8 °C (triangles, acquisitions 261–270). Each group is placed on the absolute scale with the mean ion-chamber readings of the acquisitions it contains, and the averaged buffer (69 surviving of 89 acquisitions, D0077 and D0080) is subtracted with no empirical scale factor. (c) Correlation functions at the lowest Q bin for the two high-temperature states. The dashed line is the measured correlation level of a static reference recorded in the same beamtime and read at the same Q bin; a static sample cannot relax, so that level is the instrumental contrast. Both are measurements on one axis, so no calibration is imported and none is quoted. The 33.8 °C curves run flat just below the line across four decades of delay, which is arrest; the 31.9 °C curves decay away from it, which is not. Error bars are the measured pointwise g_2 uncertainties; the y axis is capped so that a few short-delay caps clip. One key serves all three panels: its columns are the seven thermal cycles, numbered along the top and colored on a single scale, and its rows are the three temperature states, labeled at the left and distinguished by marker shape.

Data and Code Availability

Reduced data and analysis code are at `https://github.com/qzhang234/PubCode-2021-IDPP-Reversibil`

References

(1) Masters, C. L.; Simms, G.; Weinman, N. A.; Multhaup, G.; McDonald, B. L.; Beyreuther, K. Amyloid Plaque Core Protein in Alzheimer Disease and Down Syndrome. *Proceedings of the National Academy of Sciences of the United States of America* **1985**, *82*, 4245–4249.

(2) Serpell, L. C.; Berriman, J.; Jakes, R.; Goedert, M.; Crowther, R. A. Fiber Diffraction of Synthetic α-Synuclein Filaments Shows Amyloid-Like Cross-β Conformation. *Proceedings of the National Academy of Sciences of the United States of America* **2000**, *97*, 4897–4902.

(3) Cooper, G. J. S.; Willis, A. C.; Clark, A.; Turner, R. C.; Sim, R. B.; Reid, K. B. M. Purification and Characterization of a Peptide from Amyloid-Rich Pancreases of Type 2 Diabetic Patients. *Proceedings of the National Academy of Sciences of the United States of America* **1987**, *84*, 8628–8632.

(4) Pan, K.-M.; Baldwin, M.; Nguyen, J.; Gasset, M.; Serban, A.; Groth, D.; Mehlhorn, I.; Huang, Z.; Fletterick, R. J.; Cohen, F. E.; Prusiner, S. B. Conversion of Alpha-Helices into Beta-Sheets Features in the Formation of the Scrapie Prion Proteins. *Proceedings of the National Academy of Sciences of the United States of America* **1993**, *90*, 10962–10966.

(5) Radamaker, L.; Lin, Y.-H.; Annamalai, K.; Huhn, S.; Hegenbart, U.; Schönland, S. O.; Fritz, G.; Schmidt, M.; Fändrich, M. Cryo-EM Structure of a Light Chain-Derived Amyloid Fibril from a Patient with Systemic AL Amyloidosis. *Nature Communications* **2019**, *10*, 1103.

(6) Sawaya, M. R.; Sambashivan, S.; Nelson, R.; Ivanova, M. I.; Sievers, S. A.; Apostol, M. I.; Thompson, M. J.; Balbirnie, M.; Wiltzius, J. J. W.; McFarlane, H. T.; Madsen, A. Ø.; Riekel, C.; Eisenberg, D. Atomic Structures of Amyloid Cross-β Spines Reveal Varied Steric Zippers. *Nature* **2007**, *447*, 453–457.

(7) Maji, S. K.; Perrin, M. H.; Sawaya, M. R.; Jessberger, S.; Vadodaria, K.; Rissman, R. A.; Singru, P. S.; Nilsson, K. P. R.; Simon, R.; Schubert, D.; Eisenberg, D.; Rivier, J.; Sawchenko, P.; Vale, W.; Riek, R. Functional Amyloids as Natural Storage of Peptide Hormones in Pituitary Secretory Granules. *Science* **2009**, *325*, 328–332.

(8) Chapman, M. R.; Robinson, L. S.; Pinkner, J. S.; Roth, R.; Heuser, J.; Hammar, M.; Normark, S.; Hultgren, S. J. Role of Escherichia coli Curli Operons in Directing Amyloid Fiber Formation. *Science* **2002**, *295*, 851–855.

(9) Knowles, T. P. J.; Mezzenga, R. Amyloid Fibrils as Building Blocks for Natural and Artificial Functional Materials. *Advanced Materials* **2016**, *28*, 6546–6561.

(10) Pálmadóttir, T.; Getachew, J.; Ortigosa-Pascual, L.; Axell, E.; Wei, J.; Olsson, U.; Knowles, T. P. J.; Linse, S. On the Reversibility of Amyloid Fibril Formation. *Biophysics Reviews* **2025**, *6*, 011303.

(11) Kato, M. et al. Cell-free Formation of RNA Granules: Low Complexity Sequence Domains Form Dynamic Fibers within Hydrogels. *Cell* **2012**, *149*, 753–767.

(12) Hughes, M. P.; Sawaya, M. R.; Boyer, D. R.; Goldschmidt, L.; Rodriguez, J. A.; Cascio, D.; Chong, L.; Gonen, T.; Eisenberg, D. S. Atomic structures of low-complexity protein segments reveal kinked beta sheets that assemble networks. *Science* **2018**, *359*, 698–701.

(13) Luo, F.; Gui, X.; Zhou, H.; Gu, J.; Li, Y.; Liu, X.; Zhao, M.; Li, D.; Li, X.; Liu, C. Atomic structures of FUS LC domain segments reveal bases for reversible amyloid fibril formation. *Nature Structural & Molecular Biology* **2018**, *25*, 341–346.

(14) Meyer, D. E.; Chilkoti, A. Quantification of the Effects of Chain Length and Concentration on the Thermal Behavior of Elastin-like Polypeptides. *Biomacromolecules* **2004**, *5*, 846–851.

(15) Amiram, M.; Quiroz, F. G.; Callahan, D. J.; Chilkoti, A. A highly parallel method for synthesizing DNA repeats enables the discovery of 'smart' protein polymers. *Nat. Mater.* **2011**, *10*, 141–148.

(16) Quiroz, F. G.; Chilkoti, A. Sequence heuristics to encode phase behaviour in intrinsically disordered protein polymers. *Nature Materials* **2015**, *14*, 1164–1171.

(17) Varanko, A. K.; Su, J. C.; Chilkoti, A. Elastin-Like Polypeptides for Biomedical Applications. *Annu. Rev. Biomed. Eng.* **2020**, *22*, 343–369.

(18) Saha, S.; Banskota, S.; Roberts, S.; Kirmani, N.; Chilkoti, A. Engineering the architecture of elastin-like polypeptides: From unimers to hierarchical self-assembly. *Adv. Ther.* **2020**, *3*, 1900164.

(19) Sing, M. K.; Glassman, M. J.; Vronay-Ruggles, X. T.; Burghardt, W. R.; Olsen, B. D. Structure and rheology of dual-associative protein hydrogels under nonlinear shear flow. *Soft Matter* **2017**, *13*, 8511–8524.

(20) Glassman, M. J.; Olsen, B. D. Arrested Phase Separation of Elastin-like Polypeptide Solutions Yields Stiff, Thermoresponsive Gels. *Biomacromolecules* **2015**, *16*, 3762–3773.

(21) Glassman, M. J.; Avery, R. K.; Khademhosseini, A.; Olsen, B. D. Toughening of Thermoresponsive Arrested Networks of Elastin-Like Polypeptides To Engineer Cytocompatible Tissue Scaffolds. *Biomacromolecules* **2016**, *17*, 415–426.

(22) Advanced Photon Source 12-ID-B Home Page and Detector Configuration. Advanced Photon Source, Argonne National Laboratory, 2026; `https://www.aps.anl.gov/Sector-12/12-ID-B`, Accessed July 22, 2026.

(23) Xu, L.; Wang, Z.; Chen, J.; Chen, S.; Yang, W.; Ren, Y.; Zuo, X.; Zeng, J.; Wu, Q.; Sheng, H. Folded Network and Structural Transition in Molten Tin. *Nature Communications* **2022**, *13*, 126.

(24) Zheng, C. Y.; Yao, Y.; Deng, J.; Seifert, S.; Wong, A. M.; Lee, B.; Mirkin, C. A. Confined Growth of DNA-Assembled Superlattice Films. *ACS Nano* **2022**, *16*, 4813–4822.

(25) Teixeira, J. Small-Angle Scattering by Fractal Systems. *Journal of Applied Crystallography* **1988**, *21*, 781–785.

(26) Sunde, M.; Serpell, L. C.; Bartlam, M.; Fraser, P. E.; Pepys, M. B.; Blake, C. C. F. Common Core Structure of Amyloid Fibrils by Synchrotron X-Ray Diffraction. *Journal of Molecular Biology* **1997**, *273*, 729–739.